Question	Answer	Marks
8(a)	photoelectric effect	B1
8(b)(i)	$E = hf$	C1
	work function $= 6.63 \times 10^{-34} \times 8.8 \times 10^{14}$ $= 5.8 \times 10^{-19}$ J	A1
8(b)(ii)	$hf = \Phi + ½\, mv_{MAX}^2$	C1
	$6.63 \times 10^{-34} \times 11 \times 10^{14} = (5.8 \times 10^{-19}) + (½ \times 9.11 \times 10^{-31} \times v_{MAX}^2)$	C1
	$v_{MAX} = 5.7 \times 10^5$ m s^{-1}	A1
8(c)	E_{MAX} shown as zero from f = 8.0 to 8.8 and non-zero from f = 8.8 to 11	B1
	all non-zero E_{MAX} shown as a single straight line with a positive gradient	B1
	line passing through (11, 1.45)	B1

Question	Answer	Marks